CENTER FOR HISTORY OF CHEMISTRY

VLADIMIR PRELOG

Transcript of an Interview
Conducted by

Tonja Koeppel

in

Zürich, Switzerland

on

17 January 1984

Vladimir Prelog JH 3/15/96

TER FOR HISTORY OF CHEMISTRY ORAL HISTORY PROJECT

This manuscript is based on a tape-recorded interview conducted for the ACS-AIChE - University of Pennsylvania Center for History of Chemistry, the tape and the manuscript being the property of the Center. I have read the manuscript and made only minor corrections and emendations. The reader is, therefore, asked to bear in mind that this is a transcript of the spoken word rather than a literary product.

I wish to place the following condition upon the use of this interview, and I understand that the Center will enforce that condition to the fullest extent possible:

(Check One)

✓ OPEN. This manuscript may be read and the tape heard by scholars approved by the Center. The scholar pledges not to quote from, cite, or reproduce by any means this material except with the written permission of the Center.

_____ MY PERMISSION REQUIRED TO QUOTE, CITE, OR REPRODUCE. This manuscript and the tape are open to examination as above. The scholar pledges not to quote from, cite, or reproduce by any means this material except with the written permission of the Center, in which permission I must join. Upon my death this interview becomes Open.

_____ MY PERMISSION REQUIRED FOR ACCESS. I must give written permission before the manuscript or tape can be examined (other than by Center staff in the normal course of processing). Also, my permission is required to quote from, cite, or reproduce by any means. Upon my death this interview becomes Open.

V. Prelog
(Signature)

April 10, 1985
(Date)

This interview has been designated as **Free Access**.

One may view, quote from, cite, or reproduce the oral history with the permission of CHF.

***Please note*:** Users citing this interview for purposes of publication are obliged under the terms of the Chemical Heritage Foundation Oral History Program to credit CHF using the format below:

Vladimir Prelog, interview by Tonja Koeppel at the Swiss Federal Institute of Technology, Zürich, Switzerland, 17 January 1984 (Philadelphia: Chemical Heritage Foundation, Oral History Transcript # 0038).

Chemical Heritage Foundation
Oral History Program
315 Chestnut Street
Philadelphia, Pennsylvania 19106

The Chemical Heritage Foundation (CHF) serves the community of the chemical and molecular sciences, and the wider public, by treasuring the past, educating the present, and inspiring the future. CHF maintains a world-class collection of materials that document the history and heritage of the chemical and molecular sciences, technologies, and industries; encourages research in CHF collections; and carries out a program of outreach and interpretation in order to advance an understanding of the role of the chemical and molecular sciences, technologies, and industries in shaping society.

INTERVIEW: Vladimir Prelog

INTERVIEWED BY: Tonja Koeppel

PLACE: Swiss Federal Institute of Technology
Zürich, Switzerland

DATE: 17 January 1984

KOEPPEL: Professor Prelog, you were born in Sarajevo on July, 23, 1906. To what country did Sarajevo belong at that time?

PRELOG: It was part of the Austro-Hungarian monarchy. My life has been affected in many ways because Sarajevo has such a bad reputation. As you may know, Archduke Franz Ferdinand and his wife were assassinated there in 1914 and that action is considered to be the proximate cause of the first World War.

KOEPPEL: Yes.

PRELOG: I realized how Sarajevo's reputation affected me when I had to acquire a sailing permit in 1950 in the United States. While doing so, an agent of the Internal Revenue Service interviewed me. He asked me, "Professor, where are you from?"

I said, "Zürich."

He responded, "Zürich, Sweden?"

I said, "No, Zürich, Switzerland."

He was a little irritated and said to me, "You were not born, of course, in Zürich."

I answered, "No."

He asked, "Where?"

I answered, "Sarajevo."

He then said, "Oh, that's the place where all this mess started."

I was very proud that while he didn't know that Zürich is in Switzerland, he did know that Sarajevo exists.

KOEPPEL: He didn't hold it against you, did he?

PRELOG: No.

KOEPPEL: That was in 1950?

PRELOG: In 1950. Sarajevo is now famous because the Winter Olympics will be held there this year.

KOEPPEL: Yes. Well, you must have been a student in Sarajevo when the assassination occurred. Were you eight years old?

PRELOG: I was a student in elementary school at the time and

I was standing very near to the place of assassination. As a pupil I was assigned to throw flowers at the Archduke's procession. I was in the first row.

KOEPPEL: Oh, you were there!

PRELOG: The assassination occurred about two hundred meters from the place where I was standing. I didn't see it, however, because it happened around the corner.

KOEPPEL: It must have been terrible.

PRELOG: Yes, it was very bad.

KOEPPEL: There was much confusion, I guess.

PRELOG: I was, as in many other cases, a spectator of very important events.

KOEPPEL: Oh, yes, you were indeed! Two world wars. We'll get to that later. So, you went to grade school in Sarajevo.

PRELOG: Yes. We then moved to another place, Zagreb. During most of my stay in Yugoslavia I resided in the city of Zagreb, the capital of Croatia, the northern part of Yugoslavia.

KOEPPEL: What year did you move to Zagreb?

PRELOG: In 1915.

KOEPPEL: So you were in about the third grade. You entered a different school and then...

PRELOG: I then attended Realgymnasium, a kind of high school, something between elementary school and university.

KOEPPEL: Oh, I see.

PRELOG: The system is practically the same in Switzerland.

KOEPPEL: Does it take you to the university level?

PRELOG: You attend Realgymnasium for eight years and then take what we call the maturity examination. After passing this examination you may be accepted by a university. This is more or less the middle European system.

KOEPPEL: Yes. You must have studied chemistry there because I noticed that you published a paper while still in this school.

PRELOG: For two years I attended school in a smaller town, Osijek, because my father was appointed director of a high school there. I had a very, very good teacher. I helped him

to build a small chemistry laboratory. As I have mentioned on several occasions, I proved that the level of chemical literature was not very high because my paper was accepted! It was published when I was about fifteen years old.

KOEPPEL: That's wonderful! What was the paper about?

PRELOG: It was about a titration apparatus.

KOEPPEL: You actually wrote it in German!

PRELOG: I wrote it in German and it was corrected by my teacher. It was accepted by the Chemiker Zeitung.*

KOEPPEL: You were bilingual?

PRELOG: No, I could only write German, not speak it. I had studied German at school. My mother tongue was Croatian and we didn't speak German at home.

KOEPPEL: I see.

PRELOG: We read German books at home. There were many German books at home, but we didn't speak German.

KOEPPEL: Yes, well, what was your father's name?

PRELOG: My father's name was Milan. It is a Yugoslav name. He was a professor at the Gymnasium. There were different branches of Gymnasia. One emphasized Greek and Latin for people who wanted to become lawyers or even study medicine. The other branch offered mathematics, natural sciences, and modern languages. It was called Realgymnasium.

KOEPPEL: I see.

PRELOG: I had taken the scientific education, not the humanistic one.

KOEPPEL: Your father was not a scientist?

PRELOG: My father was a historian at the Gymnasium. Very soon, however, he became professor of history at the University of Zagreb. We moved back therefore to Zagreb. The last years of my schooling at this level were in Zagreb.

KOEPPEL: Did your father's being an historian influence you? You seem to have an historical bent, at least in your latest work.

PRELOG: Yes. I come from a family of historians--even my

---

*Vladimir Prelog, "Eine Titriervorrichtung," Chemiker Zeitung, 45 (1921): 736.

KOEPPEL: You may be a little bit in limbo, because you say that you forgot your mother language and then maybe you didn't get quite as deeply into your next languages.

PRELOG: Yes, you know, either you have to learn a language very young, or you have to be very talented.

KOEPPEL: I know you don't have a crystal ball, but what is your idea of the future of chemistry or of chemical education?

PRELOG: Chemistry is a very big field and it's so important because everything that happens in the world is related to it. It is so important to know in order to understand what's going on in the world. So, we have really just started and I think...

KOEPPEL: It will survive?

PRELOG: It will never stop. Consider, for example, such a complicated process like memory. Evidently, memory is something chemical. Yet we are far from understanding it. We have to do a lot of fundamental work to understand it. We don't have the necessary tools yet, but they do improve every day. Of course, Einstein said, "We live in a time of more and more perfect methods and more and more confused aims." We always have to think of what we really want to do because the possibilities now are really tremendous.

KOEPPEL: Yes, they are frightening.

PRELOG: If you take a chain of one hundred amino acids in a protein, and twenty different amino acids, you can make 10^{100} possible combinations. There are, as far as we know, 10^{86} atoms in the world and many, many proteins have hundreds or thousands of amino acids. Now, we know that we will never be able to make all of these possible isomeric proteins even though we could synthesize many of them with existing methods. If you say I want to combine a hundred amino acids in a certain sequence to a protein, you can do it. It will take years of effort, however, and require several very able coworkers. Yet, if you fail to put only a very few amino acids into the right sequence, you will not have the expected activity. Which one of these syntheses should be done? Should we spend a million dollars to prepare one of these isomers which might be of no value in the end? This is our problem. We have to be clear about the aim, about what we really want to do because we can now do much more than we were able to do before. We improve our methods every day. They become easier, faster, and more specific.

KOEPPEL: Professor Prelog, before we conclude the interview, I would like to ask whether you have any letters, papers, or

manuscripts on file. Where will your papers be stored? Are you collecting them?

PRELOG: Unfortunately, I have too many of them. It would be very difficult to sort out what is important and what is not important. I think that I shall keep the correspondence with Cahn, Ingold, and other people who contributed to the CIP system.

KOEPPEL: Where are these letters located? Are they kept right at the Institute?

PRELOG: I have them here now, but I shall perhaps deposit them into the archive of the library of the Swiss Federal Institute of Technology. The library does quite a good job collecting the correspondence and other papers of some of the members of the faculty. The library, for example, will keep all of the important correspondence that Ruzicka left. It has a special place where it keeps the correspondence and manuscripts of Thomas Mann, who was a kind of professor here during his later years. It also has the correspondence of Carl Gustav Jung and some other important members of the ETH. I think the library's staff does it professionally and very well. I don't think that I have many letters worth preserving. Some of them, of course, are more important than others, but I think that it would be a formidable task to select them and to store them. I think that I will not leave too much of the correspondence to the archives of the Swiss Federal Institute.

KOEPPEL: You will donate only what is relevant to your scientific work.

PRELOG: Only very relevant. Otherwise, very soon even the Federal Institute of Technology wouldn't have enough space.

KOEPPEL: Professor Prelog, I thank you for the interview you gave me. You have given me so much interesting information about your life, about your work and your relationship with scientists, including those in America. It has been a great experience. Thank you very much.

INDEX

adamantane	15
Adams, Roger	30
aglycone	6
alkaloid research	8, 9, 13, 14, 18, 19
American Chemical Society (ACS)	12, 26, 31
Amherst College	30
amines	32
amino acids	32
ammonium sulfite	7
animal husbandry	13
antibiotic	28
Arigoni, Duilio	26
arsenicals	8
asymmetric atoms	18
Atlantic City, New Jersey	31
Bamberger, Eugen	22
Basel, Switzerland	11, 17
Basel, University of	26
"Beilstein test"	20
Beilstein's Handbuch der Organischen Chemie	19, 20
benzopyren	8
Berichte der Deutschen Chemischen Gesellschaft	11
birth control pill	12
boars	13
Cahn, Robert Sydney	18, 19
California, University of, Berkeley	30
California, University of, Los Angeles	30
carcinogenics	8
Chemical Abstracts	20
chemical engineering	4
chemical technology	4
Chemiker Zeitung	3
Cherbuliez, Emile	11
Chicago, University of	30
chloroacetophenone	7
chromatography	31
CIBA, Ltd.	17, 27
CIBA-Geigy Co.	27
cinchona alkaloids	8, 14, 18
CIP (Cahn, Ingold, Prelog) system	19-20, 33
Columbia University	30
"computer test"	20
conformations	21
Cornell University	26, 30

Croatia 2, 9
Czechoslovakia 7

Debye-Scherrer X-ray photograph 15
diastereomer 23
diazomethane solutions 8
dihydroquinine 9
docent 9
doctoral degree 6
"Dorfaltester" 26
Dríza, Gothard 7
Dunitz, Jack D. 26

Eidgenössische Technische Hochschule (ETH) 4, 10, 17, 22, 23, 24, 33
Einstein, Albert 32
enantiomers 22, 31
Engel, Bruno 25
Ernst, Ivan 17
Eschenmoser, Albert 17, 26
European educational system 3-4
Experientia 19

Ferdinand, Archduke Franz 1,2
Fichter, Fritz 11
Fischer, Emil 18
fume hoods 8

Gäuman, Ernst 28
Geigy, Ltd. 27
Geneva, Switzerland 11
German army 9
German Chemical Society 9, 12
Germany 4, 9, 10
glycosides 6
Gosteli, Jacques 17

Hantzsch, Arthur 22
Harvard University 16, 26, 30
Heilbronner, Edgar 26
helices 21
Helmchen, Günter 20
Helvetica Chimica Acta 11
Heussler, Karl 17
Hitler, Adolf 11
Hoffman-LaRoche Co. 12
hormones 13

Illinois, University of 30
Imperial Chemical Industries (ICI) 20
Ingold, Christopher K., Sir 18, 19

Institute of Technology (Prague) 4, 9
instrumentation 28

Journal of the American Chemical Society 10
Jung, Carl Gustav 33

Karrer, Paul 11, 23, 32
Kuhn, Richard 9, 22

lachrimator 7
Lewis, Gilbert N. 30
Leuchs, Hermann 14
Librium 12
Liebig's Annalen der Chemie (Annalen) 11
liquid separation 31
London, England 19
Lukes, Rudolf 7, 15

M and P system 21
Madison, Wisconsin 30
malaria research 8, 18
Manchester, England 18, 19
Mann, Thomas 33
Marburg, University of 4
Massachusetts Institute of Technology (MIT) 30
Meyer, Victor 22
microbiology 28
musk oil 13, 14

National Science Foundation 28
natural products chemistry 6, 12, 30
nitroso urea 8
Nobel Prize 16, 18, 22, 23, 24
Nutley, New Jersey 12

Octagon House 19
organ extracts 12
Organic Chemistry Laboratory, ETH 10, 22
Osijek, Yugoslavia 2

Pauling, Linus 30
permanent waves 7
physical organic chemistry 22
polycyclic compounds 14
powder diagrams 15
Prague, Czechoslovakia 4, 9
privatdocent 10
Prelog, Milan 2, 3
Prelog, Vladimir, brother of 4
Prelog, Vladimir, grandfather of 4
Prelog, Vladimir, son of 4

proteins 32
Purdue University 30

qualitative analysis 5
quantitative analysis 5
quinine research 8, 9, 18

R and S system 20
reaction mechanisms 6
Realgymnasium 2,3
Reichstein, Tadeus 11
research funding 27-29
rhamnoconvolvulinolic acid 6
Richter, Friedrich 19
rifamycin 28
rings, chemical 15
Robinson, Sir Robert 14
Roger Adams Award (ACS) 26
Rosenkranz, George 12
rotating chairmanship 26, 27
Royal Society of Chemistry 12, 19
Ruzicka, Leopold 9, 10, 11, 12, 13, 16
22, 24, 25, 26, 27, 30, 31, 33

Sarajevo, Yugoslavia 1, 2
Scherrer, Paul 29
selenium 8
sequence rule 19
sex attractants 13
Simon, Wilhelm 31
Sputnik 29
Staedeler, Georg 22
Staudinger, Hermann 22
stereochemistry 6, 7, 15, 18, 19, 20, 21, 22
stereoformula 18
stereogenic unit 21
stereoisomer 18, 19, 20, 21
stereospecific pathways 18
Sternbach, Leo 12
steroids 12, 13
Stockholm, Sweden 23
strychnine research 14
sugar chemistry 6
Swiss-American Foundation 16
Swiss Chemical Society 11, 12
Swiss Federal Institute of Technology (ETH) 4, 10, 11, 23, 24, 33
Swiss National Science Foundation 28

Switzerland 11, 20, 23, 24, 28
Syntex Corporation 12

technical high school (technische Hochschule) 4
tellurium 8
"Thoughts After 118 Semesters of Chemistry" 4, 7
titration apparatus 3
Tollens, Bernhard 6
toxic chemical research 7, 8
truffles 13

u and l system 20
United States of America 1, 7, 10, 12, 26, 28, 29, 30, 31, 33
Urbana, Illinois 30

Valium 12
Van't Hoff, Jocobus H. 21, 22
"Verwaltungsrat" 27
vitamin B_{12} research 17
Votocek, Emil 6

Werner, Alfred 22, 23
Widmer, Sigmund 23
Wieland, Heinrich 14
Willstatter, Richard 22
Wislicenus, Johannes 22
Wisconsin, University of 30
Woodward Institute (Basel) 17
Woodward, Robert B. 14, 15-17

X-ray analysis 18

Yale University 26, 30

Zagreb, University of 3, 9, 16, 23, 24, 27
Zagreb, Yugoslavia 2, 3, 9, 10, 15
Zürich, Switzerland 1, 9, 12, 15, 17